ATHENA TECH

DATA SCIENCE RESOURCES

Basic EDA Projects

Project 1: Titanic Survival Prediction

Objective : Predict whether a passenger survived the Titanic disaster based on features like age, sex, class, and fare.

Dataset Source: Kaggle Titanic Dataset

Techniques

- Missing value imputation
- Label encoding (Sex, Embarked)
- Logistic Regression, Decision Tree, Random Forest
- Evaluation: Accuracy, ROC-AUC, Confusion Matrix

Key Features: Pclass, Sex, Age, SibSp, Parch, Fare, Embarked

Tools: Python, pandas, scikit-learn, seaborn, matplotlib

Project 2: Iris Flower Classification

Objective: Classify iris flowers into three species based on petal and sepal measurements.

Dataset Source: UCI ML Repository: Iris Dataset

Techniques:

- Exploratory Data Analysis
- Support Vector Machines, K-Nearest Neighbors, Decision Tree
- Cross-validation for performance

Key Features: SepalLength, SepalWidth, PetalLength, PetalWidth

Tools:Python, scikit-learn, seaborn, matplotlib

Project 3: Netflix Top 10 Analysis

Objective: Analyze trending content on Netflix to uncover patterns in content type, country distribution, and weekly views.

Dataset Source: Netflix Top 10 Titles

Techniques

- Time series aggregation
- Genre and country breakdown
- Barcharts, line graphs, heatmaps
- Optional: Tableau or Power BI dashboard

Key Features: Title, Week, Views, Category, Country

Tools: Python (pandas, seaborn), Tableau, or Power BI

Project 4: Google Play Store Review Analysis

Objective: Analyze user reviews to identify sentiment and common issues in apps listed on the Google Play Store.

Dataset Source: Google Play Store Apps Dataset (Kaggle)

Techniques

- Text preprocessing (cleaning, tokenizing)
- TF-IDF vectorization
- Sentiment prediction using Naive Bayes / Logistic Regression
- Word cloud for keywords

Key Features: App Name, Review Text, Sentiment, Rating

Tools: Python, NLTK, scikit-learn, WordCloud

Project 5: Airbnb Price Prediction

Objective: Predict Airbnb listing prices based on room features, location, availability, and ratings.

Dataset Source:

- Inside Airbnb Dataset
- Kaggle NYC Airbnb Open Data

Techniques

- Data cleaning and missing value handling
- Feature encoding (neighborhood, room type)
- Linear Regression or XGBoost
- Evaluation: MAE, RMSE

Key Features: Room Type, Reviews, Availability, Location, Amenities

Tools: Python, pandas, scikit-learn, seaborn
